

1. A rectangular swimming pool 50m long, 20m wide, and 5m deep is filled with water to a depth of 4m. Set up a definite integral to compute the work required to pump the water out over the top of the swimming pool.

The density of the water is 1000 kg/m^3 , and the gravitational acceleration is 9.8 m/s^2 .

Fill in the blanks in the following answer line. **Do not evaluate the definite integral.**

Answer: Work = $\int_0^4 \underline{9800 \cdot 1000 (5-y)} \underline{\quad} d \underline{y}$ J.

Formula : $W = \int_0^b \rho g A(y)(h-y) dy$



$$A(y) = 50 \cdot 20 = 1000 \text{ m}^2$$

$$h-y = 5-y$$

$$\rho = 1000, \quad g = 9.8$$

2. Find the general solution to $\frac{dy}{dx} = 10x^4 y^2$.

Write your final answer in the form of $y = f(x)$.

Special case: $y=0$: check $\begin{matrix} \text{LHS} = 0 \\ \text{RHS} = 0 \end{matrix} \Rightarrow y=0 \text{ is a solution.}$

$$\int \frac{1}{y^2} dy = \int 10 x^4 dx$$

$$-\frac{1}{y} = 2x^5 + C$$

$$\Rightarrow y = -\frac{1}{2x^5 + C}$$

Solution: $y=0$ or $y = -\frac{1}{2x^5 + C}$.